LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Installation of components and assembling of circuits on a Bread board.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Installation of components and assembling of circuits on a Bread board.
5. Specific Objectives	24.11.1.1 Trainee will be able to mount and assemble circuits on a Bread board.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Mounting and assembling circuits on a Bread board; Safety precautions regarding short circuits and component polarity; Instructor supervision during insertion and verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a breadboard in circuit prototyping?
(b) Identify three common electronic components used in basic circuits.
(c) Explain why soldering is generally avoided during breadboard assembly.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Breadboard structure and internal connectivity • Identification and selection of passive components • Correct insertion techniques for resistors and capacitors • Polarity handling for diodes and integrated circuits • Verification of connections before power application	L & Ds	PPT, IB	15 min
(b)	• Step-by-step assembly of a simple series circuit • Troubleshooting open and short circuits on breadboard • Testing circuit functionality with multimeter • Proper handling and storage of components • Documentation of circuit layout and component values	L & Ds	PPT, PDF	20 min
(c)	• Review of common assembly errors and prevention • Safety protocols for live circuit testing • Final inspection checklist for circuit integrity • Transition to PCB layout concepts • Q&A session on practical challenges	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the correct method for inserting an integrated circuit into a breadboard.
(b) How would you verify if a resistor is correctly oriented in a circuit?
(c) What steps should be taken if a circuit fails to power on after assembly?

12. Home Assignment and Follow Up	Prepare a preliminary sketch of a circuit layout for a given schematic, considering component placement and trace routing, to be used in the next lesson on PCB design.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Design and preparation of PCB layout for a given circuit.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min